

Localizing Molecular Pathology in the Human Cerebral Cortex

Scalable, layer-resolved tissue mapping from spatial transcriptomics to study neuropsychiatric and neurodegenerative disease

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The problem. The human cerebral cortex is organized into six cellular layers whose disruption underlies disorders from schizophrenia to Alzheimer’s disease. New spatial-transcriptomics technologies can finally measure gene expression *while preserving each cell’s location*, opening the door to asking *where* in the tissue molecular pathology begins. But to do so we must first assign every measured spot to its correct cortical layer – a step that today is done by slow, subjective manual annotation and does not scale to the many tissue sections a disease study requires. This project develops an accurate, automated, label-efficient method for that mapping step.

Preliminary result – already obtained on real human cortex

On the gold-standard LIBD DLPFC benchmark (12 Visium sections, 47,329 expert-annotated spots, 7 cortical domains), a pilot of the proposed spatial GNN reaches **macro-F1** 0.755 / **ARI** 0.654 under cross-section evaluation (whole tissues held out, 3 seeds), versus **XGBoost** 0.516 / 0.369 and **MLP** 0.484 / 0.375 on identical expression features – a +0.24 macro-F1 win. A graph-removal ablation collapses the identical GNN back onto the blind MLP (F1 0.49), confirming the advantage is *spatial*, not features. The win is **architecture-robust** (a graph-attention GAT encoder matches GraphSAGE at ~0.75 F1) and **generalizes to a second technology**: on single-cell *osmFISH* mouse cortex (33 genes, true region labels) the GNN reaches **macro-F1** 0.96 versus 0.54 (XGBoost) and 0.63 (MLP). The proposal below formalizes, stress-tests, and extends these results.

1 Background and Significance

The cerebral cortex is a layered structure: six neuronal laminae (L1–L6) above the white matter, each with a distinct composition of cell types and gene expression. This laminar architecture is not incidental – it organizes cortical computation, and its disruption is a recurring theme in disease:

- **Schizophrenia:** layer-specific deficits in interneurons and pyramidal-neuron signalling in the dorsolateral prefrontal cortex (DLPFC).
- **Alzheimer’s disease:** neurofibrillary tau pathology follows a *laminar* distribution, with specific layers selectively vulnerable.
- **Autism spectrum disorder:** abnormalities of cortical lamination and minicolumnar organization.

Because these changes are *spatially localized*, understanding them requires methods that respect tissue location. Spatial transcriptomics (10x Visium, MERFISH, Xenium) now provides exactly

this – genome-wide expression at thousands of spatially-registered spots per tissue section. The scientific prize is the ability to pinpoint *which layer*, and *which molecular programs within it*, change in disease.

Why it matters	Consequence
Cortical function is layer-organized	Disease changes are layer-specific, not diffuse.
Spatial omics preserves location	We can finally ask <i>where</i> pathology starts.
But spots must be assigned to layers first	Domain mapping is the rate-limiting step for every downstream analysis.

Table 1: Accurate spatial-domain mapping is the enabling step that turns spatial transcriptomics into layer-resolved disease biology.

2 The Unmet Need

To attribute a molecular change to a cortical layer, every spot must first be labelled with its domain. Today this is done in two unsatisfactory ways:

Gap 1 – Manual annotation does not scale. Expert layer annotation of a single section is slow and subjective; disease cohorts require dozens of sections across many donors, where manual labelling becomes the bottleneck.

Gap 2 – Non-spatial analysis fragments tissue. Clustering spots by expression alone ignores location, producing spatially incoherent, noisy domains because a single spot’s expression is sparse and dropout-corrupted.

What is needed is an *automated* method that is *accurate*, *label-efficient* (learns from a few annotated sections and transfers to new ones), and *spatially coherent*. The central scientific observation we exploit is simple: **a spot’s layer is a property of its spatial neighbourhood** – adjacent spots overwhelmingly share a layer – so the information missing from one noisy spot is present in the tissue around it (Fig. 1).

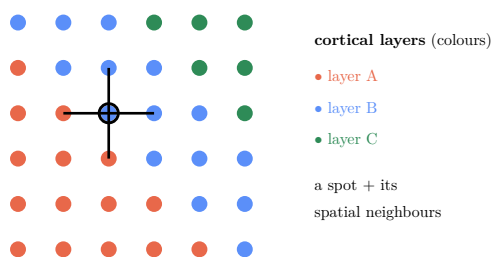


Figure 1: Layers are spatially contiguous, so a spot’s label agrees with its neighbours’. A single (noisy) spot may be ambiguous, but its neighbourhood reveals its layer – the signal an analysis must use.

3 Research Aims

Aims

Aim 1. Develop an automated method that maps spatial-transcriptomics spots to cortical layers accurately and label-efficiently.

Aim 2. Show it *transfers across tissue sections* (train on some, predict on new ones), the regime that matters for disease cohorts.

Aim 3. Verify the result is biologically grounded – recover known layer marker genes and surface candidate annotation errors for expert review.

Guiding questions. (Q1) Can layer identity be predicted accurately from spatial transcriptomics? (Q2) Does explicitly modelling tissue *neighbourhood* improve accuracy over analysing each spot in isolation, and by how much? (Q3) Do the predictions recover known cortical-layer biology?

4 Data

Resource	Content	Role
LIBD DLPFC (10x Visium)	12 sections, ~48k spots, expert manual layer annotations (L1–L6 + WM)	Real ground-truth benchmark
Controlled synthetic tissue	layers defined by neighbourhood; per-spot signal deliberately noisy	Method validation / ablation
osmFISH mouse cortex	single-cell resolution, 33 genes, ~4.8k cells, region labels	Second-technology generalization

Table 2: All data are public and de-identified post-mortem tissue. The LIBD DLPFC dataset is the field-standard ground truth for cortical-layer mapping.

5 Proposed Methodology

The neighbourhood structure in Fig. 1 motivates representing each tissue section as a **spatial graph**: nodes are spots, edges connect each spot to its k nearest spatial neighbours, and node features are the spot’s (preprocessed) gene expression. Crucially, *coordinates are used only to build the graph, never as features*, so any benefit is attributable to using tissue structure rather than memorizing absolute position.

5.1 Proposed model: a spatial graph neural network

We propose a graph neural network (GNN) that classifies each spot’s layer by pooling information across its spatial neighbourhood. With a GraphSAGE encoder over L layers of message passing,

$$h_v^{(l)} = \sigma\left(W^{(l)}\left[h_v^{(l-1)} \parallel \text{AGG}_{u \in \mathcal{N}(v)} h_u^{(l-1)}\right]\right), \quad \hat{y}_v = \text{softmax}(W_o h_v^{(L)}), \quad (1)$$

trained with cross-entropy on annotated spots. The aggregation step pools the neighbourhood, denoising the sparse per-spot signal – exactly the property Gap 2 lacks. We will compare GraphSAGE, GCN, and graph-attention variants.

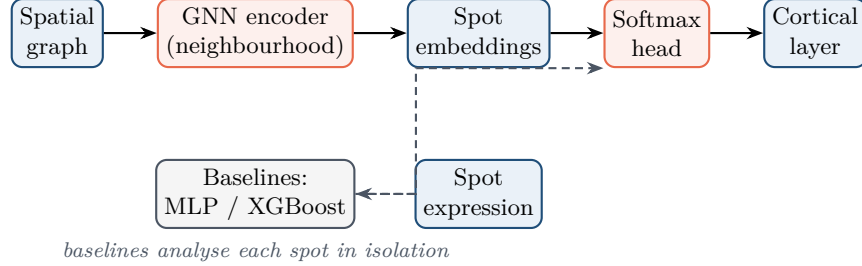


Figure 2: Proposed pipeline. The GNN (orange) pools each spot’s spatial neighbourhood; the non-spatial baselines (gray) classify each spot alone.

5.2 Baselines and rigorous evaluation

We benchmark the GNN against strong non-spatial models on the *same* expression features, so the comparison isolates the value of using tissue structure.

Model	Uses tissue structure?	Purpose
Spatial GNN (proposed)	yes (neighbourhood)	layer mapping that respects tissue
XGBoost	no	strong tabular baseline on expression
MLP	no (features only)	structure-blind control

Table 3: Baselines use the same per-spot expression features the GNN starts from; only the GNN sees the spatial graph.

Leakage-safe splits. (i) *Cross-section* – whole sections are held out for testing; since layer layouts differ per section, position cannot transfer and only neighbourhood-relative reasoning generalizes (the disease-cohort regime). (ii) *Within-section* – contiguous spatial blocks are held out. **Metrics:** accuracy, macro-F1 (layers are imbalanced), and the adjusted Rand index (ARI, the field-standard domain metric), as mean±std over 5 seeds.

6 Rigor: Ablations and Biological Interpretation

6.1 Is the spatial structure really doing the work? (falsification)

We run the identical GNN on three graphs; if the gain is genuinely spatial, destroying tissue structure must erase it.

Condition	Graph given to the model	Expected
Intact	true spatial k NN graph	high
Shuffled	random edges, same count	$\approx$ baseline
Empty	no edges	$\approx$ baseline

Table 4: The ablation makes the central claim falsifiable.

6.2 Biological interpretation → discovery

For each predicted layer we will (i) extract the genes driving predictions and confirm recovery of **known cortical-layer marker genes** used in manual annotation, and (ii) flag spots where the model confidently disagrees with the manual label – candidate *annotation errors* or genuine micro-domains for expert review. This turns an accuracy metric into biological insight.

7 Alignment with Strong-Research Criteria

Criterion	Commitment
Real, sizable public data	LIBD DLPFC: 12 sections, ~48k annotated spots.
Clear medical motivation	Layer-resolved localization of disease pathology.
Rigorous evaluation	XGBoost + MLP baselines; leakage-safe splits; 5 seeds; macro-F1 + ARI.
Interpretability → discovery	Recover layer markers; surface candidate annotation errors.
Ablations	Graph removal, message-passing depth, neighbourhood size.
Translational impact	Enables scalable, layer-resolved disease studies.

Table 5: The project is structured around the elements of strong research.

8 Expected Outcomes

A proposal states targets, not measured results.

Outcome	Target
Accurate layer mapping	High macro-F1/ARI on held-out DLPFC sections.
Value of tissue structure	Spatial model clearly > non-spatial baselines, with the gap widening across sections.
Causal check	Graph-removal ablation collapses the spatial model to the baseline.
Biological grounding	Recovered layer markers; candidate annotation errors flagged.
Honesty clause	Any narrowing of the gap is reported as-is; baselines are strong and splits leakage-safe.

Table 6: Success is a scalable, accurate, biologically-grounded layer-mapping method validated on real cortex.

9 Eight-Week Timeline

10 Reproducibility and Ethics

Implementation uses open tools (PyTorch Geometric for the model; scanpy/anndata for spatial-omics I/O). All code, configs, seeds, and a data manifest will be released. No human subjects are involved; all inputs are public, de-identified post-mortem tissue data. Predictions are research-only and hypothesis-generating and do not constitute clinical guidance.

Selected References

1. Maynard, Collado-Torres, et al. *Transcriptome-scale spatial gene expression in the human dorsolateral prefrontal cortex*. Nature Neuroscience, 2021.

Weeks	Milestones
1–2	Data pipeline (DLPFC to AnnData; spatial-graph builder); synthetic validation tissue.
3–4	Implement proposed spatial model + baselines (XGBoost, MLP); shared trainer.
5	Cross-section + within-section experiments, 5 seeds, macro-F1/ARI.
6	Ablations: graph removal, depth, neighbourhood size.
7	Biological interpretation: layer markers; candidate annotation errors.
8	Figures, tables, manuscript; reproducibility package.

Table 7: Schedule for the eight-week SRI cohort.

2. Hu et al. *SpaGCN: integrating gene expression, spatial location and histology to identify spatial domains*. Nature Methods, 2021.
3. Dong & Zhang. *STAGATE: deciphering spatial domains via a graph attention auto-encoder*. Nature Communications, 2022.
4. Hamilton, Ying, Leskovec. *Inductive Representation Learning on Large Graphs (GraphSAGE)*. NeurIPS, 2017.
5. Palla et al. *Squidpy: a scalable framework for spatial omics analysis*. Nature Methods, 2022.